Industrial Internship Report on

”Grocery Delivery Application”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UnConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was Grocery Delivery Application This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface ..... **Error! Bookmark not defined.**

2 Introduction 4

2.1 About UniConverge Technologies Pvt Ltd 5

2.2 About upskill Campus 6

2.3 Objective 7

2.4 Reference ..... 11**Error! Bookmark not defined.**

2.5 Glossary ..... 11**Error! Bookmark not defined.**

3 Problem Statement ..... 12**Error! Bookmark not defined.**

4 Existing and Proposed solution ..... 13**Error! Bookmark not defined.**

5 Proposed Design/ Model ..... 15**Error! Bookmark not defined.**

5.1 High Level Diagram (if applicable) ..... **Error! Bookmark not defined.**15

5.2 Low Level Diagram (if applicable) 15

5.3 Interfaces (if applicable) 16

6 Performance Test ..... **Error! Bookmark not defined.**17

6.1 Test Plan/ Test Cases ..... **Error! Bookmark not defined.**18

6.2 Test Procedure ..... **Error! Bookmark not defined.**18

6.3 Performance Outcome ..... **Error! Bookmark not defined.**18

7 My learnings ..... **Error! Bookmark not defined.**19

8 Future work scope ..... **Error! Bookmark not defined.**20

1. Preface

This report presents a summary of the work carried out during my **6-week internship program**, which provided me with valuable practical exposure and enhanced my technical and professional skills. Internships play a crucial role in career development as they bridge the gap between theoretical knowledge and real-world industry practices.

During this internship, I worked on the project titled **"Grocery Delivery Application"**, which focuses on developing an efficient system for online grocery ordering and delivery. The project addresses real-world challenges such as time constraints, accessibility, and the need for a convenient shopping experience.

I am grateful for the opportunity provided by **USC/UCT**, which enabled me to gain hands-on experience and understand how projects are planned and executed in a structured manner. The program was well-organized, with a clear roadmap that included learning phases, development stages, and regular progress evaluations.

Throughout this internship, I learned various technical skills related to frontend and backend development, database management, and system design. In addition, I improved my problem-solving abilities, communication skills, and understanding of teamwork in a professional environment. This experience has been highly beneficial for my overall growth and future career.

I would like to express my sincere gratitude to all those who supported and guided me during this internship. I thank my mentors, faculty members, and peers for their constant encouragement and valuable suggestions that helped me successfully complete this project.

Finally, I would like to convey a message to my juniors and peers: make the most of such opportunities, stay curious, keep learning, and focus on building practical skills, as they are essential for achieving success in your career.

2. Introduction

a. About UnConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

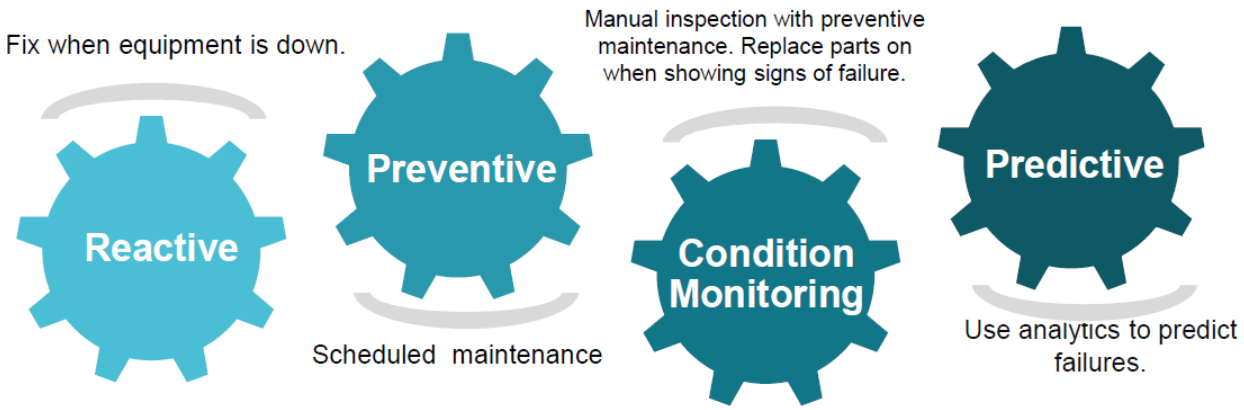
Its unique SaaS model helps users to save time, cost and money.

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iii. Predictive Maintenance

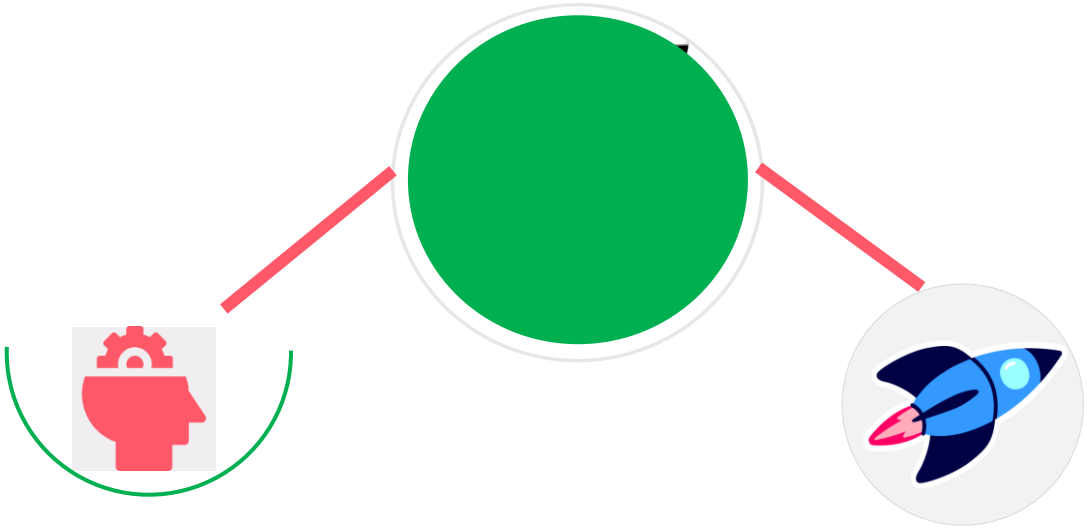
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



b. About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

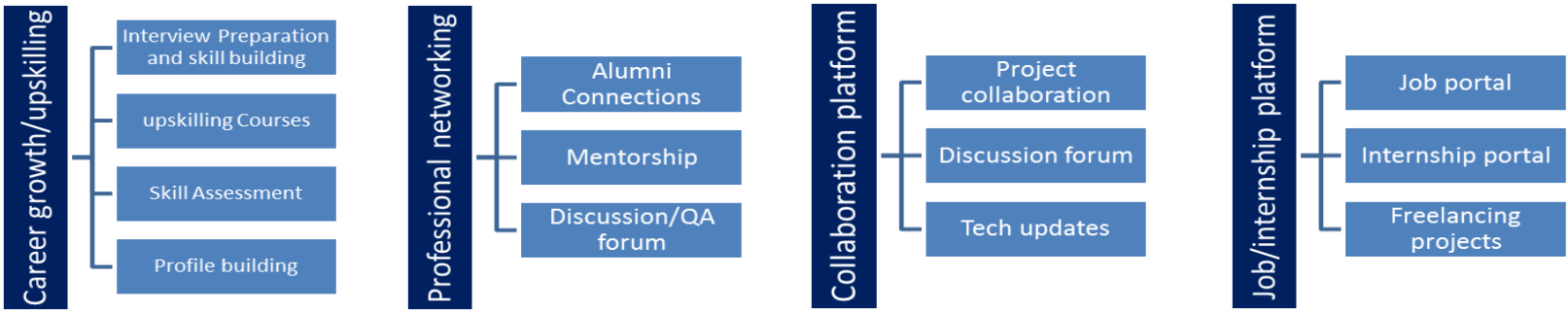
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



c. The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship Program

The objectives of this internship program were to:

- ☛ Gain practical experience of working in the industry
- ☛ Solve real-world problems using technical knowledge
- ☛ Improve job prospects and career opportunities
- ☛ Enhance understanding of the field and its real-time applications
- ☛ Develop personal skills such as communication and problem-solving

2.5 Reference

[1] Online resources and documentation for web development (HTML, CSS, JavaScript)
[2] Backend development references (Node.js / Django / Java – based on your project)
[3] Database management resources (MySQL / MongoDB)[4] Tutorials and guides from platforms like W3Schools, GeeksforGeeks

2.6 Glossary

Terms	Acronym
Application Programming Interface	API
Structured Query Language	SQL
User Interface	UI
User Experience	UX
HyperText Transfer Protocol	HTTP
Database Management System	DBMS
Software Development Life Cycle	SDLC

3. Problem Statement

In today's fast-paced lifestyle, people often face difficulty in purchasing groceries due to **time constraints, long queues, and limited store accessibility**. Traditional grocery shopping methods are time-consuming and inconvenient, especially for working individuals and elderly people.

There is a need for a system that allows users to **order groceries online easily and get them delivered to their doorstep** in a reliable and efficient manner.

The main problem is to design and develop a **Grocery Delivery Application** that enables:

- Easy browsing and selection of grocery items
- Secure and convenient order placement
- Efficient order management and delivery tracking
- Reliable and user-friendly experience
-

This solution aims to **simplify grocery shopping**, save time, and improve overall customer convenience.

4. Existing and Proposed Solution

Existing Solutions

- Many platforms provide online grocery services (e.g., basic e-commerce apps).
- Features include product browsing, cart management, and order placement.

Limitations:

- Complex user interfaces in some applications
- Limited personalization and recommendations
- High delivery time during peak hours
- Lack of real-time tracking in some systems
- Performance issues with high user traffic

Proposed Solution

- Develop a **user-friendly Grocery Delivery Application** with smooth navigation
- Provide efficient product browsing, cart, and checkout system
- Implement real-time order updates and tracking
- Ensure fast response time and optimized performance

Value Addition

- Simple and intuitive UI for better user experience
- Scalable system to handle multiple users efficiently
- Improved order accuracy and faster processing
- Future scope for AI-based recommendations and smart features

4.1 Code Submission (GitHub Link)

https://github.com/rakshitha-14/upskillcampus/blob/main/GroceryDeliveryApp.java

4.2 Report Submission (GitHub Link)

https://github.com/rakshitha-14/upskillcampus/blob/main/GroceryDeliveryReport.pdf

5. Proposed Design / Model

This section explains the overall design flow of the Grocery Delivery Application, from user interaction to final order delivery.

5.1 High Level Diagram

Description:

- User interacts with the application (Web/Mobile Interface)
- Requests are sent to the **Backend Server**
- Backend processes logic and communicates with the **Database**
- Admin manages products, orders, and users
- Delivery system handles order dispatch and tracking

Flow:
User → Frontend → Backend → Database → Order Processing → Delivery

(Add a simple diagram showing these components in blocks with arrows)

5.2 Low Level Diagram

Description:

- User Authentication Module (Login/Register)
- Product Management Module (Add/View/Search Items)
- Cart & Order Module (Add to Cart, Checkout)
- Payment Module (Transaction handling)
- Order Tracking Module (Status updates)

Flow:
Input → Validation → Processing → Database Update → Output (UI Response)

(You can draw modules connected with arrows showing internal flow)

5.3 Interfaces

- **User Interface:** Web pages for browsing products, cart, and orders
- **Admin Interface:** Dashboard for managing products and orders
- **Database Interface:** Stores user data, product details, and transactions
- **API Interface:** Communication between frontend and backend using HTTP/REST

Additional Design Elements:

- Data Flow Diagrams (DFD) for order processing
- Flowcharts for user actions (login, checkout)
- Basic protocols: HTTP/HTTPS for secure communication
- Efficient data handling using structured database queries

6. Performance Test

This section evaluates how the application performs under real-world conditions and ensures its suitability for industry use.

Identified Constraints

- **Response Time:** Fast loading of pages and order processing
- **Scalability:** Ability to handle multiple users simultaneously
- **Memory Usage:** Efficient handling of data and sessions
- **Accuracy:** Correct order details, billing, and delivery information
- **Network Dependency:** Performance under slow internet conditions

Handling Constraints in Design

- Optimized database queries to reduce response time
- Used efficient data structures and backend logic
- Implemented validation to ensure data accuracy
- Designed scalable architecture for handling multiple requests
- Minimized resource usage for better performance

6.1 Test Plan / Test Cases

- Login and authentication test
- Product search and filtering test
- Add to cart and checkout process test
- Payment processing test
- Order placement and tracking test
- Multiple users accessing the system simultaneously

6.2 Test Procedure

- Performed manual and basic automated testing
- Tested each module individually (Unit Testing)
- Conducted integration testing for end-to-end workflow
- Simulated multiple users to check system load
- Recorded response time and error rates

6.3 Performance Outcome

- Application handled multiple users with **minimal delay**
- Response time remained within acceptable limits
- Accurate order processing with no major errors
- System showed stable performance under moderate load
- Identified scope for improvement in **high-load scalability and optimization**

7. My Learnings

- This project helped me understand the complete development of a **Grocery Delivery Application**, from design to deployment.

- I gained practical knowledge in **frontend and backend technologies**, improving my technical skills.

- I learned how to integrate **databases and APIs** to manage products, users, and orders efficiently.

- The project enhanced my ability to **analyze problems and implement effective solutions**.

- I developed a better understanding of **real-world application workflows** such as order placement, tracking, and delivery systems.

- I improved my skills in **time management, teamwork, and documentation** during project development.

- These learnings will help me build a strong foundation for my career in **software development** and prepare me for real-time industry projects.

8. Future Work Scope

- Implement **real-time order tracking** using GPS for better delivery transparency.
- Integrate **online payment systems** such as UPI, credit/debit cards, and wallets.
- Add **AI-based product recommendations** to enhance user experience.
- Develop a **mobile application (Android/iOS)** for wider accessibility.
- Introduce **subscription services** for regular grocery deliveries.
- Enhance security with **advanced authentication and data protection techniques**.
- Improve system scalability to handle **large number of users and orders efficiently**.
- Add features like **chat support and feedback system** for better customer interaction.